

Steam Sterilizer / Autoclave Qualification Report

Illustrative IQ/OQ/PQ structure with mapping, heat penetration and final qualification decision.

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1. Purpose and scope

This sample report demonstrates how installation, operational and performance qualification evidence for a pre-vacuum steam sterilizer can be structured into a concise lifecycle conclusion. The sample assumes qualification of one chamber used for defined porous and hard-goods loads.

2. System and validation approach

- **System:** jacketed pre-vacuum steam sterilizer with automated cycle control and independent temperature/pressure recording.
- **Qualification stages:** IQ confirmation, OQ functional/challenge testing, PQ empty-chamber mapping and loaded heat-penetration runs.
- **Reference logic:** user requirements, approved cycle parameters, calibration status, load definition and risk-based worst-case positions.
- **Evidence:** calibrated temperature loggers, chamber pressure records, chemical/biological indicators where justified, raw-data review and deviation assessment.

Illustrative acceptance criteria shown below are fictional and must be replaced by approved site-specific criteria.

3. Qualification test matrix

Stage	Test	Evidence generated	Example decision
IQ	Installation / utilities	Equipment ID, chamber volume, piping/utilities, instrument list, software/recipe identification, calibration status	Installed condition matches approved design and controlled records
OQ	Vacuum / leak performance	Leak test, vacuum sequence, alarms/interlocks, cycle controls, recorder verification	Functions operate as specified; alarms/interlocks challenge successfully
OQ	Empty chamber distribution	Temperature mapping at representative empty-chamber locations	Defined sterilization band reached and maintained throughout exposure
PQ	Loaded heat penetration	Worst-case load, cold-spot sensors, routine configuration	All monitored load locations meet approved lethality/temperature criteria
PQ	Microbiological challenge	BI placement at justified worst-case positions where applicable	Required growth/no-growth outcome after incubation

4. Example performance results

Run	Load	Observed exposure range	Illustrative lethality	Result
PQ-01	Porous load - max configuration	132.3-133.1 °C	F0 18.7-24.3	Pass
PQ-02	Porous load - max configuration	132.1-133.0 °C	F0 18.2-23.8	Pass
PQ-03	Porous load - max configuration	132.4-133.2 °C	F0 19.1-24.8	Pass

Interpretation: All fictional runs remained within the approved exposure range. The defined cold-spot positions achieved the illustrative lethality requirement and no unresolved deviation affected the conclusion.

5. Deviations and data integrity

- One logger communication interruption was documented during pre-run setup; no exposure data were lost and the logger was replaced before execution.
- Calibration certificates were reviewed before use and remained valid through the qualification period.

- Raw electronic data, cycle printouts and executed protocol pages were reconciled before report approval.

6. Qualification conclusion

Based on the fictional evidence summarized in this sample, the sterilizer is considered qualified for the defined cycle and load configuration. Routine use remains conditional on approved operating procedures, calibrated instrumentation, controlled load patterns, deviation management and periodic requalification.

7. Example report appendices

- Executed IQ/OQ/PQ protocol
- Sensor-position diagrams
- Calibration certificates
- Raw mapping files and cycle records
- BI incubation records where applicable
- Deviation forms and impact assessments
- Traceability matrix / requirement coverage summary